1.) $y' = \frac{1}{x}$

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2.) $y' = \frac{1}{x}$

3.) $y' = \frac{3}{x}$

4.) $y' = \frac{2x}{x^2} = \frac{2}{x}$

5.) $y' = 0$

6.) $y' = 1$

7.) $y' = -x^{-2} = -\frac{1}{x^2}$

8.) $y' = \frac{3x^2}{x^3} = \frac{3}{x}$

9.) $y' = \frac{1}{2x}$

10.) $y' = 1$

11.) $y' = (-x)(-\frac{1}{x^2}) = \frac{1}{x}$

12.) $y' = \frac{2\ln x}{x}$

13.) $y' = \frac{1}{\sqrt{x}} \cdot \frac{1}{2\sqrt{x}} = \frac{1}{2x}$

14.) $y' = \ln x + 1$

15.) $y' = \frac{1}{2\sqrt{x}} \ln x + \frac{\sqrt{x}}{x}$

16.) $y' = -\frac{1}{x^2} \ln x + \frac{1}{x^2}$

17.) $y' = \frac{2x}{x^2 - 4}$

18.) $y' = \left(\frac{x+1}{x}\right) \cdot \left(\frac{1}{(x+1)^2}\right) = \frac{1}{x(x+1)}$

19.) $y' = \frac{1}{\ln x} \cdot \frac{1}{x} = \frac{1}{x \ln x}$

WITH NATURAL LOGGER RHYTHM

1.) $y^3+7y=x^3$

$3y^2\frac{dy}{dx}+7\frac{dy}{dx}=3x^2$

$\frac{dy}{dx}=\frac{3x^2}{3y^2+7}$

2.) $4x^2y-3y=x^3-1$

$8xy+4x^2\frac{dy}{dx}-3\frac{dy}{dx}=3x^2$

$\frac{dy}{dx}(4x^2-3)=3x^2-8xy$

$\frac{dy}{dx}=\frac{3x^2-8xy}{4x^2-3}$

3.) $x^2+5y^3=x+9$

$2x+15y^2\frac{dy}{dx}=1$

$\frac{dy}{dx}=\frac{1-2x}{15y^2}$

4.) $t^3+t^2y-10y^4=0$

$3t^2+2ty+t^2\frac{dy}{dt}-40y^3\frac{dy}{dt}=0$

$\frac{dy}{dt}(t^2-40y^3)=-3t^2-2ty$

$\frac{dy}{dt}=\frac{-3t^2-2ty}{t^2-40y^3}$

5.) $2x^3-3y^2=8$

$6x^2-6y\frac{dy}{dx}=0$

$\frac{dy}{dx}=\frac{x^2}{y}$

6.) $y^2+2y=2x+1$

$2y\frac{dy}{dx}+2\frac{dy}{dx}=2$

$\frac{dy}{dx}=\frac{1}{y+1}$

7.) $y^2=x^2+2x$

$2y\frac{dy}{dx}=2x+2$

$\frac{dy}{dx}=\frac{x+1}{y}$

8.) $x+\sin y=xy$

$1+\cos y\cdot\frac{dy}{dx}=y+x\frac{dy}{dx}$

$\cos y\frac{dy}{dx}-x\frac{dy}{dx}=y-1$

$\frac{dy}{dx}=\frac{y-1}{\cos y-x}$

9.) $x+\tan(xy)=5$

$1+\sec^2(xy)\cdot(y+x\frac{dy}{dx})=0$

$y+x\frac{dy}{dx}=-\frac{1}{\sec^2(xy)}$

$\frac{dy}{dx}=\frac{1}{x}\left(-\frac{1}{\sec^2(xy)}-y\right)$

10.) $2\sin x\cos y=1$

$2(\cos x\cos y+\sin x(-\sin y)\frac{dy}{dx})=0$

$\sin x(-\sin y)\frac{dy}{dx}=-\cos x\cos y$

$\frac{dy}{dx}=\cot x\cot y$

11.) $\cos y+(-x\sin y\frac{dy}{dx})=0$

$\frac{dy}{dx}=\frac{1}{x}\cot(y)$

$\left.\frac{dy}{dx}\right|_{(2,\frac{\pi}{3})}=\frac{1}{2}\cdot\frac{\sqrt{3}}{3}$

12.) $\frac{1}{2\sqrt{x}}+\frac{1}{2\sqrt{y}}\frac{dy}{dx}=0$

$\frac{dy}{dx}=-\frac{1}{2\sqrt{x}}\cdot2\sqrt{y}$

$\frac{dy}{dx}=-\frac{\sqrt{y}}{\sqrt{x}}$

$\left.\frac{dy}{dx}\right|_{(4,1)}=-\frac{1}{2}$